

Practical 3

Implement RR scheduling algorithm in Python.

Code :

```
def get_positive(msg):
    while True:
        try:
            x = int(input(msg))
            if x > 0:
                return x
            print("Invalid input! Enter a positive number.")
        except ValueError:
            print("Invalid input! Enter numbers only.")
n = get_positive("Enter number of processes: ")
bt = []
remaining = []
for i in range(n):
    x = get_positive("Enter Burst Time for P" + str(i + 1) + ": ")
    bt.append(x)
    remaining.append(x)
quantum = get_positive("Enter Time Quantum: ")
wt = [0] * n
completion = [0] * n
time = 0
round_no = 1
while True:
    unfinished = False
    print("\n----- ROUND", round_no, "-----")
    for i in range(n):
        if remaining[i] > 0:
            unfinished = True
            if remaining[i] > quantum:
                print("P" + str(i + 1), "executes for", quantum,
                    "units")
                time += quantum
                remaining[i] -= quantum
            else:
                print("P" + str(i + 1), "executes for",
                    remaining[i], "units and finishes")
                time += remaining[i]
                remaining[i] = 0
                completion[i] = time
            print("Remaining Burst Time of P" + str(i + 1) + " =",
```

```

remaining[i])
    if unfinished == False:
        break
    round_no += 1
print("\n----- FINAL RESULT -----")
print("Process\tBT\tCT\tWT\tTAT")
total_wt = 0
total_tat = 0
for i in range(n):
    tat = completion[i]
    wt[i] = tat - bt[i]
    print("P" + str(i + 1), "\t", bt[i], "\t",
          completion[i], "\t", wt[i], "\t", tat)
    total_wt += wt[i]
    total_tat += tat
print("\nAverage Waiting Time =", total_wt / n)
print("Average Turnaround Time =", total_tat / n)

```

Output :

```

Enter number of processes: 2
Enter Burst Time for P1: 3
Enter Burst Time for P2: 4
Enter Time Quantum: 3

----- ROUND 1 -----
P1 executes for 3 units and finishes
Remaining Burst Time of P1 = 0
P2 executes for 3 units
Remaining Burst Time of P2 = 1

----- ROUND 2 -----
P2 executes for 1 units and finishes
Remaining Burst Time of P2 = 0

----- ROUND 3 -----

----- FINAL RESULT -----
Process BT      CT      WT      TAT
P1      3        3        0        3
P2      4        7        3        7

Average Waiting Time = 1.5
Average Turnaround Time = 5.0

```
